POWER BI PROJECT DOCUMENTATION

BMW Car Sales Analytics Dashboard

Data | Model | DAX | Report design | Insights

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Tool: Microsoft Power BI Desktop

Data: 50,000 sales records, 2010 to 2024

Report: 3 pages, 20 DAX measures



01 Project overview

This project analyses 50,000 BMW sales records in an interactive Power BI report. It helps a sales manager see where revenue comes from, which vehicle features go with higher sales, and how regions and sales classes compare. The work has four parts: loading and checking the data, a data model, 20 DAX measures, and a three-page report with one consistent blue design.

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Total revenue (USD)	Units sold	Weighted avg price (USD)	Models	Regions

Headline figures for the full dataset, with no filters applied. Revenue is price times sales volume for every record.

Goals

- Show total revenue, units sold and growth over time at a glance.
- Compare vehicle features (fuel, transmission, color, engine, price, mileage) by sales.
- Compare regions and High and Low sales classes.
- Keep every number explainable: each measure is written down in this document.

The three report pages

Page	Question it answers	Main visuals
Overview	How much did we sell, and how is revenue changing?	5 KPI cards, revenue by fuel type, revenue and units by color, revenue by model, yearly revenue vs previous year, units by region
Product & Vehicle Strategy	Which vehicle features go with higher sales?	5 KPI cards, price gauge, units by fuel type, scatter of price vs mileage, units by color, transmission mix by region
Regional & Customer Segmentation	How do regions and sales classes compare?	4 KPI cards, revenue waterfall by region, revenue by fuel type, price by region and class, engine size trend per region

02 The dataset

The source is one Excel sheet, BMW_Car_Sales_Classification, with 50,000 rows and 11 columns.

Column	Type	What it holds
Model	Text	BMW model: 3 Series, 5 Series, 7 Series, M3, M5, X1, X3, X5, X6, i3, i8
Year	Whole number	Model year, from 2010 to 2024
Region	Text	Asia, Europe, North America, Middle East, South America, Africa
Color	Text	Red, Silver, White, Blue, Grey, Black
Fuel_Type	Text	Petrol, Diesel, Hybrid, Electric
Transmission	Text	Manual or Automatic
Engine_Size_L	Decimal	Engine size in liters, from 1.5 to 5.0
Mileage_KM	Whole number	Mileage in kilometers, up to about 200,000
Price_USD	Whole number	Price in US dollars, from 30,000 to 120,000
Sales_Volume	Whole number	Units sold for the record, from 100 to 9,999
Sales_Classification	Text	High (volume 7,000 or more) or Low (below 7,000)

Data profiling

The columns were checked for quality before building the report.

Check	Result	Meaning
Missing values	0	No blanks in any column, so no rows need to be removed
Duplicate rows	0	No repeated records
Unique values	11 models, 6 regions, 4 fuel types, 6 colors, 2 transmissions	Small lists, so slicers and legends stay readable
Sales class rule	High = volume 7,000+, Low = below 7,000	Every record falls in exactly one class

> Result: the data was already clean, so Power Query only had to load it and set the data types. Nothing was deleted or changed.

03 Data preparation and model

Data preparation in Power Query

The sheet is loaded from Excel and typed in three steps. Setting types early makes sure numbers behave as numbers in every measure and chart.

Step	What it does	Why
1. Source	Opens the Excel file and picks the BMW_Car_Sales_Classification sheet	Connects the report to the data
2. Promote headers	Uses the first row as column names	Gives every column its real name
3. Changed type	Text for Model, Region, Color, Fuel_Type, Transmission and Sales_Classification; whole number for Year, Mileage_KM, Price_USD and Sales_Volume; decimal for Engine_Size_L	Correct sums, averages and sorting

The Power Query code

```
let
    Source = Excel.Workbook(File.Contents("D:\BMW_Car_Sales_Classification.xlsx"), null, true),
    Sheet  = Source{[Item="BMW_Car_Sales_Classification", Kind="Sheet"]}[Data],
    Headers = Table.PromoteHeaders(Sheet, [PromoteAllScalars=true]),
    Typed  = Table.TransformColumnTypes(Headers, {{"Model", type text}, {"Year", Int64.Type},
             {"Region", type text}, {"Color", type text}, {"Fuel_Type", type text},
             {"Transmission", type text}, {"Engine_Size_L", type number}, {"Mileage_KM", Int64.Type},
             {"Price_USD", Int64.Type}, {"Sales_Volume", Int64.Type}, {"Sales_Classification", type text}})
in
    Typed
```

Data model

The model is one flat table, BMW_Car_Sales_Classification, which holds every attribute (model, region, fuel, color and so on) in the same row. A second table named Table only holds some of the measures.

Why one table is enough	Explanation
Same-row attributes	Every question uses columns that already sit in the same row, so no lookups are needed
No relationships	There are no relationships to maintain, so filters behave in a simple and predictable way
Small and fast	One table keeps the file light and the report quick
Year as a number	Year is a whole-number column, so year-over-year logic is written in DAX on the Year value

04 DAX measures

The report uses 20 measures. They are grouped below by purpose, with one color per group. `Data` stands for the table BMW_Car_Sales_Classification.

Sales and volume

Measure	Formula	Why it is written this way
Total Revenue	`SUMX(Data, Data[Price_USD] * Data[Sales_Volume])`	Price times units for every row, then added up. A plain SUM cannot multiply two columns
Total Sales	Same formula as Total Revenue	A second name used in one chart
Total Units Sold	`SUM(Data[Sales_Volume])`	All units sold in the selection
Total Models	`DISTINCTCOUNT(Data[Model])`	How many different models are in the selection
Weighted Average Price	`DIVIDE([Total Revenue], [Total Units Sold], 0)`	Average price per unit sold, so big sellers count more
Avg Selling Price	`AVERAGE(Data[Price_USD])`	Simple average of the listed prices
Average Mileage	`AVERAGE(Data[Mileage_KM])`	Average mileage in km
Avg Engine Size	`AVERAGE(Data[Engine_Size_L])`	Average engine size in liters

Time comparison

Measure	Formula	Why it is written this way
Prior Year Revenue	`CALCULATE([Total Revenue], FILTER(ALL(Data), Data[Year] = MAX(Data[Year]) - 1))`	ALL removes the year filter, then the year before the selected one is picked
Revenue YoY %	`DIVIDE([Total Revenue] - [Prior Year Revenue], [Prior Year Revenue], 0)`	Growth against the previous year; DIVIDE avoids errors when the denominator is zero

Shares and rankings

Measure	Formula	Why it is written this way
% Revenue Contribution	`DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL(Data)), 0)`	Revenue of the selection as a share of the grand total
Eco Share %	`DIVIDE(CALCULATE([Total Revenue], Data[Fuel_Type] IN {"Hybrid","Electric"}), [Total Revenue], 0)`	How much revenue comes from Hybrid and Electric cars
Diesel Share %	`DIVIDE(CALCULATE([Total Revenue], Data[Fuel_Type] = "Diesel"), [Total Revenue], 0)`	How much revenue comes from Diesel cars
Top Model	`TOPN(1, VALUES(Data[Model]), [Total Revenue])`	TOPN keeps the one model with the highest revenue, shown on a card

Sales classes and regions

Measure	Formula	Why it is written this way
High Volume Units	`CALCULATE([Total Units Sold], Data[Sales_Classification] = "High")`	Units from High records

Measure	Formula	Why it is written this way
Low Volume Units	`CALCULATE([Total Units Sold], Data[Sales_Classification] = "Low")`	Units from Low records
High Class Txns	`CALCULATE(COUNTROWS(Data), Data[Sales_Classification] = "High")`	Number of High records (rows)
NA Revenue	`CALCULATE([Total Revenue], Data[Region] = "North America")`	Revenue of one region; CALCULATE changes the region filter
EU Revenue	`CALCULATE([Total Revenue], Data[Region] = "Europe")`	Same idea for Europe
Asia Revenue	`CALCULATE([Total Revenue], Data[Region] = "Asia")`	Same idea for Asia

Key DAX ideas

Idea	Used in	In simple words
SUMX	Total Revenue	Goes row by row, multiplies, then adds up
CALCULATE	Region revenues, class units, shares	Changes the filter for one measure only
ALL	Prior Year Revenue, % Revenue Contribution	Ignores the current filters to reach a total or another year
DIVIDE	Every ratio	Safe division: no error when the denominator is zero
TOPN	Top Model	Picks the best row from a list

05 Report design

One blue palette and one layout are used on every page, so the report feels like a single product. The colors below are the ones used in the report charts, cards, and buttons.

Deep navy
#0A3C54
Card text, darkest chart series

Main blue
#1A6D94
Most-used chart color

Sky blue
#399FCF
Second series, highlights

Steel blue
#6B99B0
Supporting chart series

Teal blue
#336A84
Supporting chart series

Slate
#445158
Neutral chart series

Button blue
#3E85C5
Action buttons

Light grey
#EFEFEF
Page navigator

Off white
#F7F7F7
Background shapes

Mid grey
#CCCCCC
Borders and lines

Layout and navigation

- Same grid on every page: a navigation panel on the left, a title bar on top, a row of KPI cards under it, and charts below.
- Page navigator: a list on the left that jumps between the three pages.
- Clear Filter buttons: each page has a button tied to a bookmark (Overview, Product & Vehicle Strategy, Regional & Customer Segmentation) that resets every filter.
- Cross-filtering: clicking a bar, slice or point filters the other visuals on the same page.
- Logo and filter icon: a brand image at the top of the panel and a small filter icon on the clear-filter button.

Chart types and why they were chosen

Chart	Used for	Why
KPI card	Headline numbers	One number each, readable in a second
Donut	Fuel type, color, region splits	Shows the share of each part of a small list
Line chart	Revenue by year, engine size by region	Shows change over time
Clustered bar / column	Models, regions	Easy comparison between categories
100% stacked column	Transmission by region	Compares the mix, not the size
Scatter	Price vs mileage vs units	Shows how two numbers relate, with size for volume
Funnel	Units by fuel type	Ranks a short list from largest to smallest
Gauge	Average selling price	Shows one value against its range
Waterfall	Revenue by region	Shows how each region adds to the total

06 The report pages

6.1 Overview

Question: How much did we sell, and how is revenue changing over time?

KPI cards

Card	What it shows
Total Revenue	Price times units, all records
Total Units Sold	All units sold
Revenue YoY %	Growth against the previous year
Average Mileage	Average mileage in km
Total Models	Number of models in the selection

Visuals

Visual	Fields used	What it shows
Donut	Fuel_Type, Total Revenue	Revenue split by fuel type
Combo (column + line)	Color, Total Sales, Total Units Sold	Revenue and units for each color
Clustered bar	Model, Sales_Classification, Total Revenue	Revenue by model, split into High and Low
Line chart	Year, Total Revenue, Prior Year Revenue	Each year against the year before
Clustered column	Region, Total Units Sold	Units sold in each region

How to read it

- Start with the five cards for the big picture, then use the line chart to see growth year by year.
- The bar chart ranks models; the two colors inside each bar show how much comes from High and Low records.
- Click a region column to see the whole page for that region only.

6.2 Product & Vehicle Strategy

Question: Which vehicle features go with higher sales?

KPI cards

Card	What it shows
Avg Selling Price	Simple average of Price_USD
Eco Share %	Revenue share of Hybrid and Electric
Diesel Share %	Revenue share of Diesel
Avg Engine Size	Average engine size in liters
Top Model	Model with the highest revenue

Visuals

Visual	Fields used	What it shows
Gauge	Avg Selling Price	Average price against its range
Funnel	Fuel_Type, Total Units Sold	Units ranked by fuel type
Scatter	Mileage_KM, Price_USD, Total Units Sold, Sales_Classification	How price and mileage relate to volume, colored by class
Donut	Color, Total Units Sold	Units by color
100% stacked column	Region, Transmission, Total Units Sold	Manual and Automatic share inside each region

How to read it

- The Eco Share and Diesel Share cards show how the fuel mix changes when you pick a region or year.
- In the scatter chart, look for clusters: do cheap or low-mileage cars sit in the High class?
- The 100% columns all reach the top, so compare the split between the two colors, not the height.

6.3 Regional & Customer Segmentation

Question: How do regions and sales classes compare?

KPI cards

Card	What it shows
NA Revenue	North America revenue
Asia Revenue	Asia revenue
EU Revenue	Europe revenue
High Class Txns	Number of High-class records

Visuals

Visual	Fields used	What it shows
Waterfall	Region, Total Revenue	How each region adds to total revenue
Donut	Fuel_Type, Total Revenue	Revenue by fuel type
Clustered bar	Region, Sales_Classification, Avg Selling Price	Average price per region and class
Line chart	Year, Region, Avg Engine Size	Engine size trend for each region

How to read it

- The three region cards compare the biggest markets; use the waterfall to see all six regions.
- Use the fuel-type donut after clicking a region to see its fuel mix.
- The line chart helps spot regions that move toward larger or smaller engines over time.

07 Key insights

All figures come from the full dataset with no filters. Each finding has the numbers behind it.

Revenue is spread evenly across models

Evidence: 7 Series leads with 9.4% of revenue, 3 Series has 9.3%, i8 9.3%, and the lowest model (M3) still has 8.8%.

Meaning: No single model carries the business; all eleven are close.

Asia is the largest region, by a small margin

Evidence: Asia 17.1%, Europe 16.8%, North America 16.7%, Middle East 16.7%, South America 16.4%, Africa 16.4%.

Meaning: Regions are almost equal, so differences are small and should be read carefully.

Half of revenue comes from eco-friendly cars

Evidence: Hybrid and Electric together make 50.3% of revenue. Hybrid is the top fuel type with 25.4%; Diesel has 24.6%.

Meaning: Hybrid and Electric are as strong as Petrol and Diesel together.

High-volume records matter more than their number suggests

Evidence: High records are 30.5% of rows but bring 51.1% of revenue.

Meaning: A smaller group of large-volume sales produces half of the money.

Manual and Automatic are balanced

Evidence: Manual 50.2% of revenue, Automatic 49.8%.

Meaning: Transmission type does not separate high from low sales.

Revenue is stable over time

Evidence: Yearly revenue stays between 1.22 trillion (2023, lowest) and 1.34 trillion (2022, highest). 2024 is 7.2% above 2023.

Meaning: There is no strong trend; changes from year to year are small.

08 Limitations and next steps

Limitations

• The data looks synthetic: values are spread almost evenly across categories, so differences between models, regions, and fuel types are small. The insights are for practice, not for real BMW decisions.

• Revenue is price times sales volume for each record, which gives a very large total (about 19 trillion USD). Treat the numbers as relative, not as real revenue.

• There is no date table, so time analysis works at the year level only.

• The Power Query source points to a local path (`D:\BMW_Car_Sales_Classification.xlsx`), so the path must be changed on another computer.

Possible improvements

Improvement	Why it helps
Add cards for Middle East, South America and Africa	Only three of the six regions have a revenue card now
Make Top Model return text: MAXX(TOPN(1, VALUES(Model), [Total Revenue]), [Model])	Safer if two models tie, and always returns a name
Remove the duplicate measure (Total Sales and Total Revenue are identical)	Keeps the model tidy and avoids confusion
Add a date table if month-level data becomes available	Allows monthly trends and standard time-intelligence functions
Use a relative or parameterized file path	Lets anyone refresh the report without editing the source

09 Conclusion

The project covers the full BI workflow on a clean 50,000-row dataset: checking the data, building a simple model, writing 20 explainable DAX measures, and designing a three-page interactive report in one consistent blue design. A manager can see the big picture on the Overview page, then move to vehicle features or regions to find the reason behind a number.